Pre-work · The Heart

The Heart and the Cardiac Conduction System. Read both Part A chapters first. Almost every question on this material is a pathway question, so the order is the content.

Module 3 · 2 of 4 | Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read**
   Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize**
   Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw**
   Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply**
   Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve**
   Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the heart and conduction chapters into mini visuals. Six chunks. Two are chains, and those two carry most of the exam weight.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A patient has a murmur heard when the ventricles contract, and imaging shows a torn papillary muscle in the left ventricle. Name the valve involved and explain the sound.

Answer. The mitral valve.

Reasoning. Work from the structure named. Papillary muscles exist only in the ventricles and anchor only the atrioventricular valves, so a semilunar valve is ruled out immediately. The left ventricle narrows it to the mitral valve. When the ventricle contracts, the pressure behind that valve is enormous, and the chordae tendineae held by that papillary muscle are the only thing stopping the cusps from being pushed back into the atrium. Torn anchor, cusps flip, blood slips backward, and the turbulence is the murmur. Notice the sound is timed to ventricular contraction, which is itself a clue.

takes longer than three minutes you are copying instead of organizing.

1 LADDER

The heart wall and pericardium

Five rungs, from fibrous pericardium on the outside down to endocardium on the inside. Write beside the epicardium rung the second name it goes by, because that is the one people miss.

2 GRID

The four chambers

Chamber down the side. Receives from, pumps to, wall thickness, and one internal feature across the top. Four rows.

3 GRID

The four valves

Valve down the side. Type, number of cusps, location, and what it prevents across the top. Add a fifth column: has chordae, yes or no.

4 CHAIN

The pathway of blood, nine steps

Nine boxes with arrows, right atrium through to the aorta. Colour the oxygen-poor half one way and the oxygen-rich half the other, and mark exactly where the switch happens.

5 CHAIN

The conduction pathway, six steps

SA node through to Purkinje fibers, arrows the whole way. Mark where the delay happens and write beside it what the delay buys.

DRAWING 1

The heart, frontal section, with flow arrows

BRAIN DUMP FIRST

Two minutes. Every chamber, valve, and great vessel, and the direction blood moves.

Draw the heart cut front to back so all four chambers show. Add the interatrial and interventricular septa, all four valves in position, chordae tendineae and papillary muscles in both ventricles, and the five great vessels attached where they belong. Now draw arrows through the whole path and shade oxygen-poor and oxygen-rich blood differently.

In your notebook, head the page **M3-1 The heart, frontal section, with flow arrows** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can start at any point in your drawing and say what comes next, forward or backward, without looking at the notes.

Set A · Name it

1. The visceral layer of the serous pericardium is also called the ________________________.
2. The vessel returning oxygen-rich blood to the left atrium is the ________________________.
3. The remnant of the fetal foramen ovale in the adult atrium is the ________________________.
4. The only electrical bridge between the atria and the ventricles is the ________________________.
5. Name the two coronary arteries and one main branch of each.

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Set B · Predict it

1. The anterior interventricular artery is blocked. Predict which region of the heart is damaged and why.

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2. Fluid collects in the pericardial cavity. Predict the effect on filling, and name the condition.

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3. The delay at the AV node is lost. Predict what happens to ventricular filling.

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4. A right coronary artery occlusion causes a rhythm problem as well as muscle damage. Explain why, anatomically.

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Set C · Tell it apart

1. Distinguish the atrioventricular from the semilunar valves by structure and by what holds each shut.

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2. Explain why the left ventricle wall is thicker than the right, using the circuit each one drives.

6 **HUB**

Coronary circulation

Aorta at the hub. Two arteries off it, then their named branches, then the matching cardiac veins running back to the coronary sinus. Draw the vein alongside the artery it travels with.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

The conduction system on a heart outline

BRAIN DUMP FIRST

One minute. Every component in order, and where each one sits.

Draw a plain heart outline. Mark the SA node in the right atrial wall near the superior vena cava, the AV node in the atrial floor, the AV bundle in the interventricular septum, both bundle branches, and the Purkinje fibers spreading through the ventricle walls. Draw the fibrous skeleton as a band between atria and ventricles and mark the one place the signal can cross.

In your notebook, head the page **M3-2 The conduction system on a heart outline** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain from your drawing why a block at the AV bundle disconnects the atria from the ventricles entirely.

3. A transplanted heart still beats though its nerves were cut. Explain what that proves.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

An ECG tracing lined up with the pathway

BRAIN DUMP FIRST

One minute. The three deflections and the conduction event behind each.

Draw one heartbeat of an ECG: P wave, QRS complex, T wave. Under each deflection write the conduction event that produces it and the chamber action that follows. Then draw a second tracing in a different colour with the P waves marching independently of the QRS complexes, and label what that is.

In your notebook, head the page **M3-3 An ECG tracing lined up with the pathway** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can point at any part of the tracing and name what the conduction system is doing at that moment.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 The heart wall and pericardium LADDER

2 The four chambers GRID

3 The four valves GRID

4 **The pathway of blood, nine steps** CHAIN

5 **The conduction pathway, six steps** CHAIN

6 **Coronary circulation** HUB